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The Editor-in-Chief
Engineering Applications of Artificial Intelligence
Elsevier

Dear Editor,

We submit the manuscript "Timestamp-Alignment Leakage in Drive-Test Handover Prediction: Quantification, Mitigation, and Blind Second-Operator Validation" for consideration as an original research article in Engineering Applications of Artificial Intelligence.

Machine learning and artificial intelligence predictors are increasingly trained on the one-hertz tabular exports of commercial drive-test tools for mission-critical telemetry and cellular network operations. The manuscript shows that an unexamined property of such exports, rather than of the network or the learning algorithm, can produce most of the reported predictive skill. In four campaigns on a commercial LTE network, the export row stamped just before a handover command already carries the post-handover serving cell in 76% of handovers. Ignoring this temporal alignment defect inflates one-second Area Under the Precision-Recall Curve (AUPRC) about four-fold, and the inflation comes mainly from serving-cell radio measurements rather than from cell identifiers. We support the mechanism with a native-rate public corpus from three further operators, and we replicate the main findings on a capture from a second operator recorded after the pipeline was frozen. We also show that the splitting protocol changes which learner appears best. The manuscript closes with a short audit and a reporting checklist that apply to any machine-learning predictor built on aggregated measurement exports.

We believe the work fits the journal's scope on engineering applications of artificial intelligence, real-world data-driven methodologies, and trustworthy machine-learning deployment in mission-critical infrastructure.

The work is based on the authors' B.Sc. thesis at the Islamic University of Technology, and its source repository is public. The manuscript has not been published elsewhere and is not under consideration by another journal. All authors have approved the manuscript and agree with its submission. The code, data and the tables behind every reported number are archived at https://doi.org/10.5281/zenodo.23479562. One of the authors (M. T. Kawser) is a co-author of a public dataset mentioned, but not used for any result, in the manuscript; this is stated in the declaration of competing interest.

Thank you for considering our manuscript.

Sincerely,

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